

Learning with Unknown Input Observers for Robust Nonlinear Estimation

A hybrid, certified estimator for systems with
unknown dynamics

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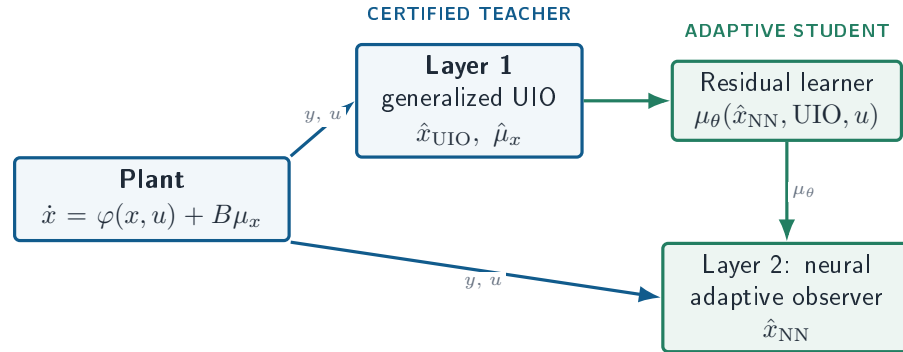
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Memory cue: Brief welcome, name, and title.

Good morning everyone, and thank you for joining us. I am Quang Huy Nguyen, and today I will present our work on learning with unknown input observers for robust nonlinear estimation.

Global architecture: certified estimation guides learning



Global architecture: certified estimation guides learning

Memory cue: State the core idea through the complete architecture.

Let me begin with the core idea and the global view. Our physical model is nonlinear, and part of its dynamics is unknown. Instead of asking one estimator to do everything, we combine a certified model-based observer with an adaptive learning layer.

From left to right, the measured output and known input first feed the generalized UIO. This certified teacher estimates both the state $\hat{x}_{\text{UIO}}$ and the unknown input $\hat{\mu}_x$. These bounded, physics-based signals supervise the residual learner. The learned residual μ_θ is then injected into Layer 2, a neural adaptive observer that propagates its own state estimate $\hat{x}_{\text{NN}}$ and remains anchored to the measurements.

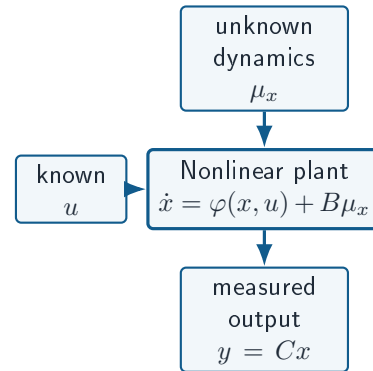
This is the roadmap for the talk: first I will construct and certify Layer 1; then I will return to this architecture to explain the Layer-2 observer and its online learning loop; finally, I will show the vehicle results.

Why estimation becomes difficult with unknown dynamics

Nonlinear plant with an unknown input

$$\begin{aligned}\dot{x} &= \varphi(x, u) + B\mu_x, & x &\in \mathbb{R}^{n_x}, \\ y &= Cx, & \mu_x &\in \mathbb{R}^{n_\mu}.\end{aligned}$$

- ▶ $\varphi(x, u)$: known nonlinear physics;
- ▶ μ_x : unmodelled dynamics, disturbances, or force residuals;
- ▶ only y is measured and μ_x must be inferred online.



Goal: estimate x and infer μ_x .

Why estimation becomes difficult with unknown dynamics

Memory cue: Known physics plus an unknown input. Estimate both the state and the residual.

Let us start from the model on this slide. The state evolves according to the known nonlinear physics, written as $\varphi(x, u)$, plus an unknown input $B\mu_x$. We measure only $y = Cx$.

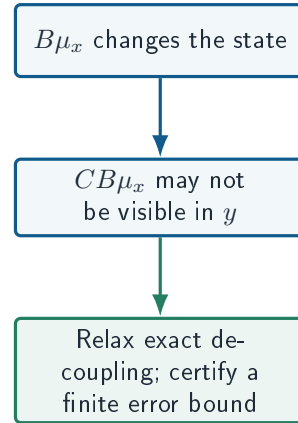
The term μ_x collects what the nominal model does not explain: unmodelled dynamics, disturbances, or force residuals. The difficulty is that this term affects the state, but it is not directly measured. Therefore, a good estimator should reconstruct the physical state robustly and provide a useful estimate of the unknown input. Classical unknown-input observer theory is useful, but it also reveals its main limitation.

The classical UIO condition is often too restrictive

Direct unknown-input decoupling requires

$$\text{rank}(CB) = \text{rank}(B).$$

- ▶ This guarantees that the output is informative enough to isolate μ_x .
- ▶ In real vehicle and robotic systems, CB can be rank deficient.
- ▶ Enforcing exact decoupling can then rule out an otherwise useful observer.



The classical UIO condition is often too restrictive

Memory cue: Classical UIO needs rank. Real systems often do not give it.

A conventional unknown-input observer relies on the rank condition shown here: the rank of CB must equal the rank of B . Intuitively, the measured output must contain enough information to separate the effect of the unknown input from the state dynamics.

When this condition holds, an observer can exactly decouple the unknown input and obtain exponential convergence. However, this assumption is often too strong in practice. A sensor set may not see every direction in which an unknown tire force, disturbance, or model mismatch acts. In that case, CB can be rank deficient. Instead of discarding the observer design, our approach relaxes exact decoupling and replaces it with a quantified estimation bound.

Regularization converts a rank obstacle into a bounded disturbance

$$\begin{aligned}\dot{x} + E_{\delta_1} \dot{\mu}_x &= \varphi(x, u) + B\mu_x + E\omega_{\delta_1}, \\ y &= Cx + D_{\delta_2}\mu_x + D\omega_{\delta_2}, \\ E_{\delta_1} &= \delta_1 E, \quad D_{\delta_2} = \delta_2 D, \\ \omega_{\delta_1} &= \delta_1 \dot{\mu}_x, \quad \omega_{\delta_2} = -\delta_2 \mu_x.\end{aligned}$$

- ▶ The transformed model is equivalent to the original plant.
- ▶ The tunable δ_1, δ_2 restore observer design freedom.
- ▶ The price is explicit: ω_δ enters the performance analysis.

Certified consequence

$$\|\tilde{\zeta}\|_{\mathcal{H}^1} \leq \lambda_\delta \|\mu_x\|_{\mathcal{H}^1} + \beta \|\xi_0\|.$$

Exact convergence is recovered when the classical rank condition holds.

Regularization converts a rank obstacle into a bounded disturbance

Memory cue: Do not force exact decoupling; make the remaining effect measurable and bounded.

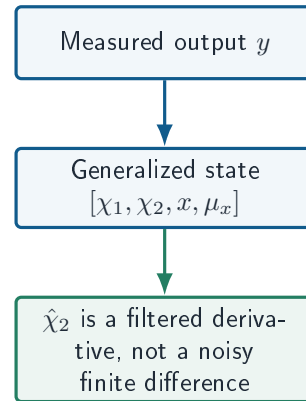
The key step is a regularized reformulation of the original system. We introduce the tunable quantities δ_1 and δ_2 , and write an equivalent model in which the derivative of the unknown input and the unknown input itself appear through the disturbance vector ω_δ .

This does not change the underlying plant. The regularization creates enough freedom to construct an observer even when the direct rank condition is not available. The cost is explicit: the unknown-input effect enters the performance analysis. The inequality on the right says that the augmented-state estimation error is bounded in the $\mathcal{H}^1$ norm by the unknown dynamics and by the initial estimation error. If the classical rank condition is available, the regularization can be removed and exact convergence is recovered.

Output derivatives create a generalized unknown-input observer

Introduce $\chi_1 = y$ and $\chi_2 = \dot{y}$ in the augmented state $\zeta = [\chi_1^\top \ \chi_2^\top \ x^\top \ \mu_x^\top]^\top$:

$$\begin{aligned}\dot{\chi}_1 &= \chi_2, \\ \dot{\chi}_2 &= C\bar{\varphi}_x(x, u) + C\frac{\partial\varphi}{\partial x}(x, u)B\mu_x + CB\dot{\mu}_x, \\ \dot{x} &= \varphi(x, u) + B\mu_x, \\ y_\chi &= \chi_1 + C_\chi\chi_2 + Cx.\end{aligned}$$



Output derivatives create a generalized unknown-input observer

Memory cue: Augment with output derivatives, but estimate them through the observer.

To strengthen the information available to the observer, we introduce the output and its derivative as auxiliary variables: $\chi_1 = y$ and $\chi_2 = \dot{y}$. These variables are included with the state and unknown input in one generalized augmented state.

The derivative channel brings additional information about the nonlinear dynamics and the unknown input. We do not take a noisy numerical derivative of the measurement. The generalized observer estimates χ_2 internally, so $\hat{\chi}_2$ behaves as a filtered output derivative. This gives more information without the usual finite-difference noise amplification. With this model, we can synthesize the first certified estimation layer.

Layer 1: LMI-certified generalized UIO

Observer realization

$$\begin{aligned}\dot{\eta} &= P_{\zeta} \varphi_{\zeta}(\hat{\zeta}, u) + K(y - \mathcal{H}\hat{\zeta}), \\ \hat{\zeta} &= \eta + Q_{\zeta} y, \quad P_{\zeta} \mathcal{E} + Q_{\zeta} \mathcal{H} = I.\end{aligned}$$

$$K = \mathcal{P}^{-1} Q^{\top}.$$

For every vertex j of the nonlinear embedding, solve

$$\begin{bmatrix} \Xi_j & \Gamma_j & \Theta_j \\ \Gamma_j^{\top} & -\lambda I_{2n_{\mu}} & \Gamma_j^{\top} \\ \Theta_j^{\top} & \Gamma_j & -\epsilon^{-1} \mathcal{P} \end{bmatrix} \prec 0.$$

- ▶ offline convex synthesis;
- ▶ online observer with a quantified robustness level λ_{δ} .

Guarantee

Feasibility yields an $\mathcal{H}^1$ estimation bound for the state and unknown input.

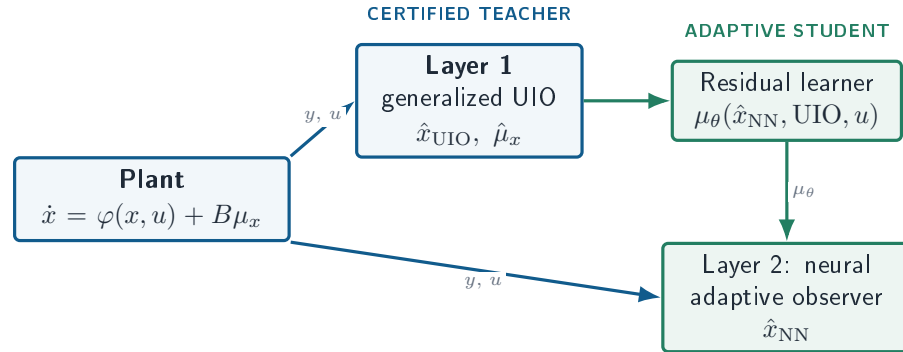
Layer 1: LMI-certified generalized UIO

Memory cue: Offline LMI, online robust observer.

This slide gives the first-layer observer. Its estimated augmented state $\hat{\zeta}$ contains the output-related variables, the physical state, and the unknown input estimate. The matrices P_{ζ} and Q_{ζ} enforce the structural decoupling relation, while the gain K is computed from the LMI solution.

The LMI is solved offline at the vertices of the nonlinear embedding, which turns the gain design into a convex feasibility problem. Once it is feasible, Layer 1 has an $\mathcal{H}^1$ estimation guarantee. It is therefore not only an estimator, but also a source of supervisory signals whose robustness can be quantified before deployment. This reliability is exactly what we need before introducing learning.

A certified teacher supervises learning



Teacher: bounded, physics-based supervisory signals

Student: adaptive residual learning

A certified teacher supervises learning

Memory cue: Layer 1 teaches; Layer 2 learns the part that the fixed model misses.

This diagram summarizes the hybrid architecture. The plant provides the measured output and receives the known input. Layer 1, the generalized UIO, estimates both the state $\hat{x}_{\text{UIO}}$ and the unknown residual $\hat{\mu}_x$.

These quantities are passed to the neural approximation. They act as a teacher signal: they are not assumed to be perfect, but they are bounded and physically meaningful because they come from the certified observer. Layer 2 is the neural adaptive observer. It uses the neural approximation together with measurement feedback to refine the state estimate. The first layer protects learning with a robust baseline; the second captures state-dependent effects that a fixed observer model cannot represent as well.

Layer 2: a neural observer estimates its own state

State-dependent residual model

$$\mu_{\theta}(\hat{x}_{\text{NN}}) = \mathcal{N}_{\theta}(\hat{x}_{\text{NN}}, \hat{x}_{\text{UIO}}, \hat{\mu}_x, u).$$

Layer-2 observer dynamics

$$\begin{aligned}\dot{\hat{x}}_{\text{NN}} &= \varphi(\hat{x}_{\text{NN}}, u) + B\mu_{\theta}(\hat{x}_{\text{NN}}) + L_{\text{NN}}(y - y_{\text{NN}}), \\ y_{\text{NN}} &= C\hat{x}_{\text{NN}}.\end{aligned}$$

The UIO supplies supervisory inputs; it does not replace the independent dynamic state $\hat{x}_{\text{NN}} \in \mathbb{R}^{n_x}$.

Three contributions to $\dot{\hat{x}}_{\text{NN}}$

- ▶ known nonlinear physics φ ;
- ▶ learned unknown dynamics $B\mu_{\theta}$;
- ▶ measurement innovation $L_{\text{NN}}(y - y_{\text{NN}})$.

Certified error channel

With $\epsilon_{\text{NN}} = x - \hat{x}_{\text{NN}}$ and $\Delta_x = \mu_x - \mu_{\theta}$,

$$\begin{aligned}\dot{\epsilon}_{\text{NN}} &= (\nabla_x^{\varphi} - L_{\text{NN}}C)\epsilon_{\text{NN}} + B\Delta_x, \\ \|\epsilon_{\text{NN}}\|_{\mathcal{L}_2} &\leq \lambda_{\text{NN}} \|\Delta_x\|_{\mathcal{L}_2} + \beta_{\text{NN}} \|\epsilon_{\text{NN}}(0)\|.\end{aligned}$$

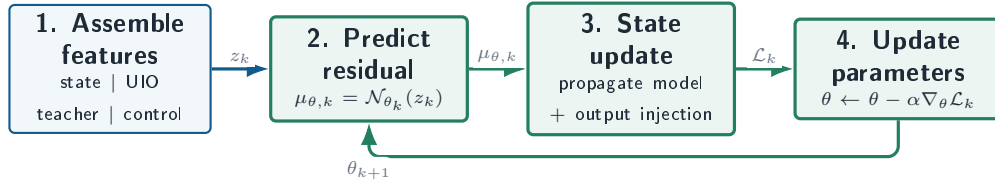
Layer 2: a neural observer estimates its own state

Memory cue: Layer 2 is an observer with its own state, not only a neural correction block.

The neural model takes the current Layer-2 estimate together with the UIO state estimate, the UIO unknown-input estimate, and the known input. It predicts the state-dependent residual μ_{θ} .

Crucially, Layer 2 propagates its own state $\hat{x}_{\text{NN}}$. Its dynamics combine the known nonlinear physics, the learned residual through $B\mu_{\theta}$, and the measurement innovation through $L_{\text{NN}}(y - y_{\text{NN}})$. Subtracting this observer from the plant gives the error dynamics on the right. The remaining approximation error Δ_x is the input to the certified $\mathcal{L}_2$ bound. The next slide explains how online learning reduces this mismatch.

Layer 2: online parameter adaptation



Composite objective at sample k

$$\mathcal{L}_k = \|\hat{x}_{\text{NN},k} - x_{\text{ref},k}\|_{T_{\text{ref}}}^2 + \|y_k - C\hat{x}_{\text{NN},k}\|_{T_y}^2 + \|\hat{x}_{\text{NN},k} - \hat{x}_{\text{UIO},k}\|_{T_{\text{UIO}}}^2 + \lambda \|\mu_{\theta,k} - \hat{\mu}_{x,k}\|^2.$$

The four terms respectively promote reference tracking, measurement consistency, agreement with the UIO, and residual matching to the teacher proxy.

Features and parameter update

$$\begin{aligned} z_k &= \text{col}(\hat{x}_{\text{NN},k}, \hat{x}_{\text{UIO},k}, \hat{\mu}_{x,k}, u_k), \\ \mu_{\theta,k} &= \mathcal{N}_{\theta_k}(z_k), \\ \theta_{k+1} &= \theta_k - \alpha_k \nabla_{\theta} \mathcal{L}_k, \quad \alpha_k > 0. \end{aligned}$$

The LMI-designed observer gain L_{NN} remains fixed while the network parameters adapt online to reduce Δ_x .

Layer 2: online parameter adaptation

Memory cue: Learn the residual, keep output feedback, and bound the remaining mismatch.

The feature vector collects the current neural-state estimate, the UIO state estimate, the UIO unknown-input estimate, and the known input. The network produces $\mu_{\theta,k}$, which enters the state observer shown on the previous slide.

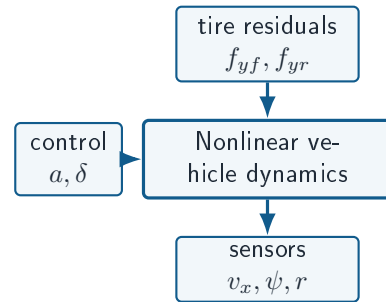
The four loss terms promote reference tracking, measurement consistency, agreement with the certified UIO teacher, and residual matching to the teacher proxy. Gradient descent updates only the neural parameters. The LMI-designed innovation gain remains fixed, so online adaptation reduces the residual mismatch without changing the certified observer structure.

Case study: nonlinear vehicle with unknown tire forces

Vehicle state, input, and measurement

$$x = [v_x \ v_y \ \psi \ r]^T, \quad u = [a \ \delta]^T, \\ y = [v_x \ \psi \ r]^T, \quad \mu_x = [f_{yf} \ f_{yr}]^T.$$

- ▶ Lateral velocity v_y is not directly measured.
- ▶ Front/rear lateral tire-force residuals represent the unknown input.
- ▶ A layered controller provides the manoeuvring input used by both estimators.



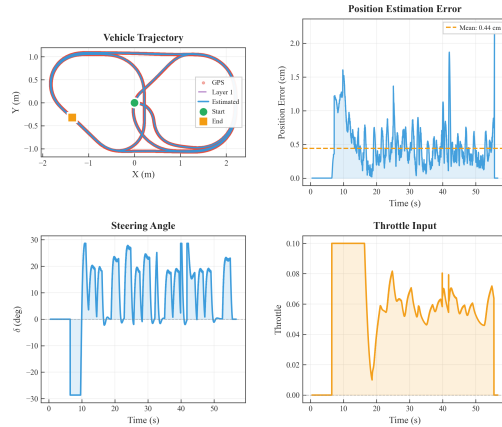
Case study: nonlinear vehicle with unknown tire forces

Memory cue: Unmeasured lateral velocity and unknown front–rear tire-force residuals.

We evaluate the method on a nonlinear vehicle model. The state contains longitudinal velocity, lateral velocity, yaw angle, and yaw rate. The known inputs are acceleration and steering angle. The measured output contains longitudinal velocity, yaw angle, and yaw rate, so the lateral velocity v_y is not measured directly.

The unknown input is composed of the front and rear lateral tire-force residuals. These residuals represent the difference between the nominal tire model and the effective forces needed to explain the motion. The vehicle setting illustrates the broader problem: use known physics where it is reliable, then estimate and learn the structured dynamics that are missing.

Trajectory tracking: centimetre-scale position error



Observed outcome

The estimated trajectory closely follows the GPS reference during the manoeuvre.

Reported metric

Mean position error: **0.44 cm**.

The lower panel confirms feasible steering and acceleration commands.

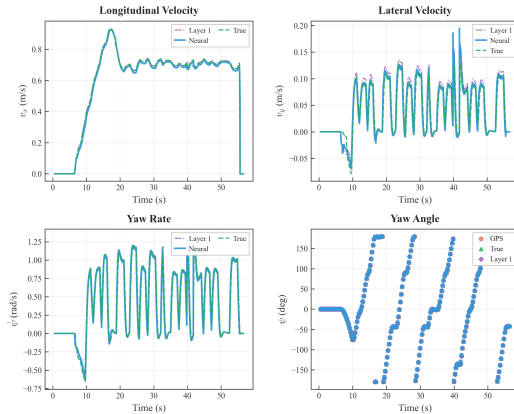
Trajectory tracking: centimetre-scale position error

Memory cue: First practical result: the estimated path follows GPS closely.

First, consider the trajectory result. In the upper-left plot, the estimated trajectory closely follows the GPS reference throughout the manoeuvre. This means that the combined estimator remains consistent at the position level, even while it handles nonlinear dynamics and unknown tire-force effects.

The upper-right plot reports the position error. The mean position error is 0.44 centimetres in this simulation. The key point is not only the small average value; it is also the absence of a sustained drift. The lower plots show steering and throttle inputs, confirming that this performance is achieved during a manoeuvre with changing control inputs rather than in a static operating condition.

State estimation: the unmeasured lateral velocity is reconstructed



What matters

- ▶ Measured states remain consistent with their sensor channels.
- ▶ The estimator supplies v_y , which is absent from y .
- ▶ The learned observer refines transient performance beyond its certified teacher.

State estimation: the unmeasured lateral velocity is reconstructed

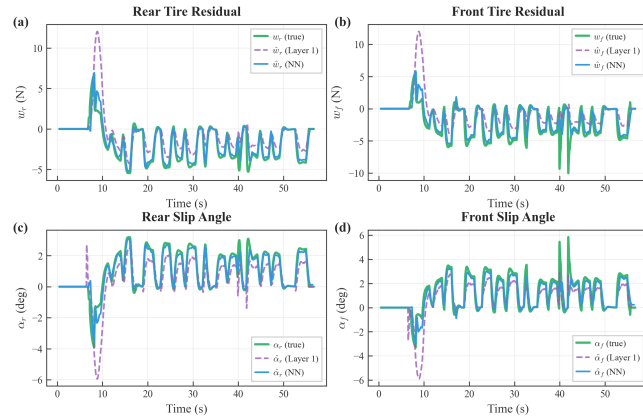
Memory cue: Measured states stay consistent; the estimator supplies the state that sensors do not give.

This slide shows the individual state estimates. The measured channels remain consistent with their sensor signals. More importantly, the estimator reconstructs the lateral velocity v_y , which is not part of the measurement vector.

The lateral-velocity plot is a useful test of the hybrid approach because its dynamics are affected by uncertain tire forces. The UIO provides the stable baseline and the neural layer refines the transient behaviour. The yaw-rate and yaw-angle plots convey the same overall message: the observer follows nonlinear vehicle motion while maintaining output consistency. This reconstruction is then reflected in the tire-force residual estimates.

Residual learning improves tire-force reconstruction

Tire Force and Slip Angle Estimation



Interpretation

- ▶ Layer 1 gives a bounded, physically meaningful residual estimate.
- ▶ Layer 2 converges faster and tracks tire-force residuals more closely.
- ▶ The learned map captures state-dependent residual dynamics.

Residual learning improves tire-force reconstruction

Memory cue: The teacher is bounded; the student is faster and closer.

Finally, these plots compare the reconstructed tire-force residuals and slip angles. Layer 1 already gives a bounded and physically meaningful residual estimate. That is valuable by itself, because it turns an unknown modelling effect into an interpretable signal.

Layer 2 improves this estimate. The neural adaptive observer converges faster and tracks the front and rear tire-force residuals more closely over the manoeuvre. This is what we expect from a state-dependent residual model: the neural map can adapt to nonlinear effects that a fixed observer gain and nominal physics cannot capture completely. It refines, rather than replaces, the certified model-based estimate.

Key takeaways

1. **Relaxed UIO design:** regularization and output derivatives remove reliance on exact unknown-input decoupling.
2. **Certified robustness:** LMs provide an explicit $\mathcal{H}^1$ bound for the generalized observer and an $\mathcal{L}_2$ bound for the neural observer.
3. **Reliable learning:** the UIO supplies stable supervisory signals; the neural observer learns the residual dynamics online.
4. **Vehicle evidence:** accurate trajectory tracking, recovery of the unmeasured lateral state, and improved tire-force estimates.

Central message

Certification and learning are complementary: the observer makes adaptation trustworthy, and adaptation makes the observer more accurate.

Key takeaways

Memory cue: Relax the rank condition, certify the baseline, then learn the residual.

Let me conclude with four takeaways. First, the generalized unknown-input observer relaxes the need for exact unknown-input decoupling by using regularization and output derivatives. Second, the method keeps analytical guarantees: the first layer has an $\mathcal{H}^1$ robustness bound, and the neural observer has an $\mathcal{L}_2$ performance bound.

Third, the learning stage is guided by a certified teacher rather than being trained in isolation. This makes the learning signals more reliable and preserves physical interpretability. Finally, in the nonlinear vehicle example, the architecture achieves accurate trajectory tracking, reconstructs the unmeasured lateral velocity, and improves tire-force residual estimation. Certification and learning should work together: the observer makes adaptation trustworthy, and adaptation makes the observer more accurate. Thank you for your attention. I would be happy to take your questions.